

Metaphony in Substance Free Logical Phonology

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Abstract

We provide an overview of Substance Free Logical Phonology (LP) and apply this model to a few case studies of synchronic metaphony and similar processes. We adopt Inkelas’s idea of exceptionality as prespecification, along with the implementation of feature filling rules based on a partial operation called unification to show that some cases of metaphony that are treated as morphophonological and/or morphosyntactic can instead be handled by LP in narrow phonology. We also explain how Inkelas’s suggestion for rule targets extends to rule triggers, as well.

1 Theoretical background

We provide an overview of Substance Free Logical Phonology (henceforth LP) and apply this model to a few case studies of synchronic metaphony and similar processes. In LP, the purview of phonology is quite narrow, and all intrasegmental processes (i.e., featural changes affecting a single segment), are derived through combinations of two kind of rules: features can be added to a segment by rules via the operation of unification, and features can be removed from a segment by rules via subtraction. In this paper, these are the only two kinds of operations we consider. Insertion and deletion of full *segments*, for example, requires additional machinery, as do syllabification, stress, and other phenomena not addressed here. We are not able to fully enumerate our views on the specification of rule environments and the nature of locality, a topic of great importance in the study of metaphony and similar phenomena, though Dabbous et al. (2021) provide some preliminary work on this matter.

Given the fact that we reduce all intrasegmental rules to these two types, unification-based rules and subtraction-based rules, our goal is to show that various phonological processes that go by names like assimilation, palatalization, vowel harmony, metaphony, and so on can be reduced to computations with the same operations and abstract representations. On one hand, this leads us to the perspective that labels such as assimilation or metaphony are what Chomsky (2000:8) calls “taxonomic artifacts, useful for informal description perhaps but with no theoretical standing.” On the other hand, it is clear that a volume like the present one is methodologically extremely useful: the focus on metaphony serves a similar purpose to micro-variation studies in syntax that help us “to abstract from the welter of descriptive complexity certain general principles governing computation that would allow the rules of a particular language to be given in

very simple forms” (op. cit.:122). We use the name *Logical Phonology* in order to reiterate the formal, substance-free nature of phonology as we conceive it, and also because we think that much of phonology can be done using very simple mathematico-logical notions borrowed set theory. The toolkit we use here is presented in more detail by Bale and Reiss (2018) and Reiss (2021), and it provides us with a means of developing a constrained theory of *possible phonological rules*.

1.1 Substance-freeness

We assume an innate set of binary phonological features transducing from the output of the grammar to the performance systems involved in speech production and perception (Chomsky and Halle 1965, Volenec and Reiss 2018, Reiss and Volenec 2022). The model is “substance-free” in the sense that the motor and perceptual correlates of a feature are not accessible to the phonological computational system. In LP, a process turning +LATERAL to –LATERAL before a +ROUND vowel has the same status as one turning +HIGH to –HIGH before a +NASAL segment, regardless of the relative phonetic plausibility of the two processes, their relative typological frequency, their presence or absence in children’s speech, or any other factors discussed in the literature under the rubric of *markedness*. *Substance-free* can in some sense be understood as *markedness-free*. The two hypothetical processes above are represented differently because they are formulated with different features and coefficients, but for grammatical computations, we assume features are arbitrary symbols as discussed by Hale and Reiss (2008).

As a point of comparison, consider assimilation, which Hyman (1975) defines as the insertion of a feature specification αF on a segment immediately adjacent to another segment specified αF .

(1) Assimilation schemata (after Hyman 1975:159):

$$\begin{array}{l} X \rightarrow \{\alpha F\} \quad / \quad _ [\alpha F] \\ X \rightarrow \{\alpha F\} \quad / \quad [\alpha F] _ \end{array}$$

In autosegmental phonology, assimilation is instead conceptualized as the sharing of feature specifications via the insertion of association lines, and phonological tiers provide a more general notion of adjacency, but the basic notion remains the same. LP also uses “Greek letter” coefficient variables to express identity or agreement (and obligatory non-identity or disagreement) between the values for a feature on various segments, but what LP denies is that there is any need to recognize or formalize notions like assimilation (or dissimilation) as less marked than other processes. Yes, there are rules that encode agreement between segments, for example, rules that make an obstruent agree in voicing with an obstruent to its immediate right, or that delete a glide between identical vowels, or that raise mid vowels before high vowels, and LP can easily express such rules. However, there are also rules which raise mid vowels before low vowels, before nasals, or before a word boundary. The following principle expresses the LP position in general terms.

(2) Substance-freeness of structural change: Features changed by rule application need not be present in the rule’s structural environment.

This principle is a claim that proposed phonological rules need not “make sense” in featural terms. Informally, it holds that the *whatness* of a rule, what feature is being added to a segment, is

logically independent of the *whereness*, the characterization of the triggering environment. This in turn echoes Chomsky and Halle’s (1968:428) claim that “the phonological component requires wide latitude in the freedom to change features”. Note that the principle is not itself an axiom of LP, but rather something that is *not* part of the theory.

1.2 Segments as sets of features

For the kinds of phenomena we examine in this paper, it is sufficient to treat segments as sets of valued features. It is not necessary to concern ourselves with the association of these sets to timing slots as one must be when dealing with *segment* insertion and deletion rules, for example. Thus individual segments, just like structural changes, are denoted in the curly brackets of set theory, as in (3).¹

(3) Segments as sets of features:

$$/i/ = \left\{ \begin{array}{l} +\text{SYLLABIC} \\ -\text{BACK} \\ -\text{ROUND} \\ +\text{HIGH} \\ -\text{LOW} \\ +\text{ATR} \end{array} \right\} \quad /e/ = \left\{ \begin{array}{l} +\text{SYLLABIC} \\ -\text{BACK} \\ -\text{ROUND} \\ -\text{HIGH} \\ -\text{LOW} \\ +\text{ATR} \end{array} \right\}$$

LP permits underspecified segments, so a set like (4) is also a legitimate representation for a segment; it denotes a vowel which lacks a specification for the feature HIGH but which is otherwise featurally identical to /i, e/. Following common practice, we use capital letters, like /I/, to abbreviate underspecified segments.

$$(4) \ /I/ = \left\{ \begin{array}{l} +\text{SYLLABIC} \\ -\text{BACK} \\ -\text{ROUND} \\ -\text{LOW} \\ +\text{ATR} \end{array} \right\}$$

In set-theoretic terms, this vowel /I/ is the intersection of the vowels /i/ and /e/, each of which is a set of features. Underspecification is another example of something that is *not* part of the theory; there is simply no axiom in LP that requires segments to be *complete*; i.e., fully specified for all features.

It is crucial to note a notational convention of LP by which representation (4) is distinguished from representation (5), which is intended as a *partial description*. We use square brackets to denote natural classes; (5) is the intensional characterization of a set of segments where each segment is set of feature specifications.

¹We assume these features for concreteness, but the exact featural structure is not at issue.

(5) The natural class containing /i, e, I/:

$$\left[\begin{array}{l} +\text{SYLLABIC} \\ -\text{BACK} \\ -\text{ROUND} \\ -\text{LOW} \\ +\text{ATR} \end{array} \right]$$

This representation is an abbreviation for unwieldy set expression (6) below.

$$(6) X = \{x : x \supseteq \{+\text{SYLLABIC}, -\text{BACK}, -\text{ROUND}, -\text{LOW}, +\text{ATR}\}\}$$

In prose, X is the set of segments whose members are all and only the segments whose featural specifications are supersets of the feature specification set $\{+\text{SYLLABIC}, -\text{BACK}, -\text{ROUND}, -\text{LOW}, +\text{ATR}\}$. We assume that rule targets are natural classes (e.g., “apply the rule to any labial”) as are the rule environments (e.g., “apply the rule before front vowels”) and we use square brackets to denote these two components. In contrast, the change specified by the rule does not refer to a set of segments, but to a set of featural changes, and therefore regular set-theoretic curly brackets are used to specify the structural change. We will encounter many examples of rules stated using these conventions below.

Given the representations for /i, e, I/ proposed above, it is possible to specify singleton natural classes of fully specified segments using the square bracket notation, attending to the distinction between natural classes and segments.

(7) Natural class versus segment:

$$\begin{array}{cc} \text{singleton class } \{i\} & \text{segment } /i/ \\ \left[\begin{array}{l} +\text{SYLLABIC} \\ -\text{BACK} \\ -\text{ROUND} \\ +\text{HIGH} \\ -\text{LOW} \\ +\text{ATR} \end{array} \right] & \neq \left\{ \begin{array}{l} +\text{SYLLABIC} \\ -\text{BACK} \\ -\text{ROUND} \\ +\text{HIGH} \\ -\text{LOW} \\ +\text{ATR} \end{array} \right\} \end{array}$$

Similarly, one can denote the singleton natural class $\{e\}$ by changing $+\text{HIGH}$ to $-\text{HIGH}$ in (7). However, it is *not* possible in this system to construct a natural class containing only the segment /I/. Informally, this is because the relevant set is a proper superset of the sets corresponding to /i/ and to /e/. In general, there is no way to construct a natural class containing a segment p underspecified relative to another segment q ; that is, where $p \subseteq q$. /I/ contains the features of the natural class representation in (5), but so do /i, e/.

Bale et al. (2020) note one interesting corollary of this formalization: it is no longer necessary to provide a feature SEGMENT so that $[\text{+SEGMENT}]$ can refer to the natural class of all segments in a language. Rather, this class can be referred to by an empty pair of square brackets, $[\]$, because one axiom of set theory is that every set—and here, every segment—is a superset of the empty set. In other words, an empty pair of square brackets has the following meaning.

$$(8) X = \{x : x \supseteq \{\}\}$$

This is the set of all segments.

1.3 Unification and subtraction

The focus of much work in the LP corpus has been the development and application of a mechanism to overcome the apparent representational limitation pointed out above. How can feature-filling processes be formulated if it is not possible to refer to underspecified segments?² The approach taken in LP is to introduce changes in the featural content of segments via unification, denoted by the $\sqcup$ operator. The unification of a segment (set) and the set of features corresponding to the structural change is equivalent to the set theoretic *union* of the two sets, but only if the union of the two sets is consistent, if it contains no conflicting feature values, like +F and –F. If the union of the two sets is not consistent, unification *fails* (or equivalently, its output is *undefined*). Unlike union, which always yields an output set for any two input sets, and is called a total operation, unification is called a partial operation, a property that LP exploits.

A rule that unifies the segment /i/ with the set {+HIGH} applies vacuously, because the set denoted by /i/ already contains +HIGH—both union and unification are vacuous. However, the same rule applied to the segment /e/, specified as –HIGH, results in unification failure because the *union* of the two sets includes both +HIGH and –HIGH. We assume, as part of the overall interpretative procedure for rule application, that when unification fails in a phonological rule application, the target segment (the first of the two arguments) remains unchanged. Thus, the unification *rule* applied to /e/ with {+HIGH} vacuously yields /e/. Finally, the unification of /I/ with {+HIGH} is equivalent to the union of /I/ with {+HIGH}, and yields /i/. The rule *appears* to have only targeted /I/, but this is an illusion: it non-vacuously targets /I/, and, vacuously targets /i, e/, though for different reasons in each case.

Reiss (2021) gives copious examples of this mechanism, as well as a second type of rule involving set difference or set subtraction, denoted here by the $\setminus$ operator. Subtraction followed by unification yields feature-changing processes: subtraction can remove features and unification can add them. As in ordinary set theory, $A \setminus B$ denotes a set containing all and only those elements of A which are not elements of B ; for example, $\{a, b, c\} \setminus \{a, x, y\} = \{b, c\}$. Subtraction can also be vacuous, as in $\{a, b, c\} \setminus \{x, y\} = \{a, b, c\}$. Together, rules based on unification and subtraction cover a wide range of phonological patterns.

²LP is not the first body of literature to consider this issue. For example, Lees (1961:12–14) discusses it in the context of Turkish vowel harmony and proposes a principle that rules may not refer to the absence of a feature. It is also discussed by Lightner (1971:236). Hayes' (1986:331) Linking Constraint ("association lines in structural descriptions are interpreted as exhaustive") is an attempt to address this problem in autosegmental phonology; see also proposals by Kiparsky (1985) and Archangeli (1988). Further problems and a proposed solution are discussed at length by Reiss (2003). Bonet (1995:631, fn. 28) notes that the problem extends beyond phonology, to a problem in clitic selection, but does not offer a solution.

2 Method and metaphony

Below we put this feature logic to use in the study of metaphony, but we note, once again, that metaphony is a pre-theoretic, phenomenological label that has no formal status in our theory. Our LP assumptions will allow us to develop the following four basic themes.

(9) Themes of LP's approach to metaphony:

- i. Apparent lexical and/or morphosyntactic conditioning of phonological targets can be handled by a judicious choice of underlying forms, especially the use of underspecification.
- ii. Apparent lexical and/or morphosyntactic conditioning of phonological environments can be handled with underspecification along similar lines.
- iii. Unification provides an elegant solution to the formalization of feature-filling processes without the need for rule diacritics.
- iv. Feature-changing processes reflect subtraction followed by unification.

These notions are all present, in some form or another, in the literature, but LP provides a simple unifying framework to express them, and metaphony provides the corpus with which we can illustrate them. Before proceeding, we note that some patterns described as metaphony reflect completed sound changes that correspond to no synchronic rule. LP is a theory of synchronic grammar and thus it has nothing to say about language change.

2.1 Favoring narrow phonology

For examples of metaphony that are potentially amenable to synchronic analysis, some, classified as *morphophonology*, seem to require reference to the identity of particular lexical items, and others, usually classified as *suppletion*, make reference to morphosyntactic structure, as well as to particular lexical items. Other productive cases of metaphony can be analyzed straightforwardly as purely phonological processes. For clarity we say that this last type of metaphony is as generated by the *narrow phonology*. Our goal in this paper is to show that our restrictive model of the narrow phonology can handle at least some metaphony patterns previously attributed to morphophonology or suppletion.

We adopt the standard—but usually implicit—position that a narrow phonological analysis is preferable, *ceteris paribus*, to a morphophonological or morphosyntactic (suppletion-based) analysis. We interpret the *ceteris paribus* condition to mean that a purely phonological solution is to be favored if it can be achieved without enriching the ontological commitments of the narrow phonology, by exploiting only a restricted, independently motivated phonological toolkit.

Under this strategy, it is desirable to attempt to bring all instances of synchronic metaphony under the umbrella of the narrow phonology. Methodologically, this is the only sensible strategy,

since the alternative would license us to posit that, say, the alternation of the plural in English *dog*[z] vs. *cat*[s] might involve morphophonology or even suppletion. Aside from missing the generalization about phonological conditioning by stem-final consonants, we would also miss the fact that the same generalizations apply to the possessive suffix in *dog*'[z] vs. *cat*'[s] and in the verbal agreement ending in *jog*[z] vs. *walk*[s].

Similarly, consider a child acquiring a language like German, in which word-final voiced obstruents devoice across the board and without exception. Most linguists assume that this child acquires a narrow phonological rule of word-final devoicing rather than a morphophonological rule (perhaps one in which the alternating roots are stored in a list mentioned in the rule) or a system of suppletive allomorphy involving every alternating root. The attested data is compatible with all three kinds of analysis (although *wug*-tests with nonce forms might provide confirmatory evidence for the narrow phonology solution). What principle leads us to favor a purely phonological solutions in general, even in the absence of productivity tests? Just as the analyst has to prioritize narrow phonological solutions, we assume that the learner is epistemically bounded (in the sense of Fodor 1980:33f.) to first seek a narrow phonological grammar before entertaining morphophonological or morphosyntactic conditioning. The language acquisition device resorts to morphophonology or suppletion only when the resources of the innate narrow phonological faculty are unable to handle an alternation.³ Of course, the child has an advantage over the analyst—the child knows what phonological Universal Grammar consists of. We, the linguists, in contrast, have to rely on Ockham's Razor and adopt an initially restrictive model.⁴

Another way to say the same thing is to imagine the learner as programmed, insofar as the innate narrow phonology allows, to induce a one-to-one mapping in the lexicon between sound (i.e., the phonological representation of morphemes) and meaning. Only when the innately determined representational and computational resources fail to generate such a one-to-one mapping, does the learner resort to morphophonology or suppletion. This position—that phonology defines the extent to which the one-to-one correspondence holds between sound and meaning—may not be correct, but it is at least an explicit hypothesis that can be explored, and it provides a principled justification for prioritizing phonological solutions rather than delegating these problems to morphology. As we show in section 4, even highly-articulated morphophonological theories stand to benefit from our approach, since we show that some of the patterns for which they have accepted responsibility can, in fact, be reassigned to the narrow phonology under LP assumptions.

³To make this concrete, it is probably the case that there is no constraint ranking in traditional versions of Optimality Theoretic that will derive *go* and *went* from a single phonological root form. Since the phonology can't handle this pattern, under OT assumptions, it must not be phonological, under OT assumptions. The same reasoning applies, in principle, to any model that is explicit enough to define its range of possible phonological grammars.

⁴This line of argument demonstrates the problem with the application of what Gopnik (2003) calls "theory theory" to cognitive development. "Theory theory" proposes a strong parallel between the learning that takes place in development and the work of scientists. However, scientists can run experiments and make use of negative evidence, which children acquiring language are believed to largely ignore; and children's genetic endowment constrains the hypothesis space for language, whereas scientists have to discover that, say, that syntactic processes require reference to notions like c-command.

2.2 Inalterability as prespecification

We take inspiration from work by Inkelas and colleagues (e.g., Inkelas and Cho 1993, Inkelas 1995, Inkelas and Orgun 1995, Inkelas et al. 1997) on apparent exceptions to phonological processes analyzed under the slogan “inalterability as prespecification”. In brief, the idea is that given a feature-filling rule that supplies segments with a specification of +F, the rule will modify segments lacking a specification for F. However, segments that appear immune to the rule—i.e., that are not altered by the rule—should be understood as “prespecified” for –F. In some sense, these prespecified segments are not exceptional; they just are not subject to change by a feature-filling rule. As discussed above, unification—and its vacuous application via unification failure—provides a formal mechanism to implement feature filling. If the tools of narrow phonology include operations like unification and subtraction, and if underspecified segments are permitted, then we predict a child will converge on grammars like those we propose below.

3 Baztan Basque low vowel assimilation

Before turning to a more traditional case of metaphony, let us first consider an example of vowel-vowel interaction to illustrate Inkelas’s insight. Hualde (1991) discusses a process called Low Vowel Assimilation (henceforth, LVA) present in several dialects of Basque. In general, LVA is attributed to “a rule that raises a low vowel to /e/ after a high vowel, with or without any intervening consonants” (op. cit.:23). Some examples, common to all dialects with LVA, are given below; /-a/ is the singular definite suffix.

(10) Low Vowel Assimilation (loc. cit.):

/mutil-a/	[multile]	‘the boy’
/mendi-a/	[mendie]	‘the mountain’
/egun-a/	[eyune]	‘the day’

Our discussion here focuses on Hualde’s account of the Baztan dialect (§2.2). In this dialect, the process affects vowels within words and across certain word boundaries, as shown in the following examples.

(11) Auxiliary verbs with raised vowel after high vowel (op. cit.:29–30.):

a.	torri de	‘he has come’	cf.	gan da	‘he has gone’
b.	gain de	‘he will go’	cf.	lorriko da	‘he will come’
c.	torri gera	‘we have come’	cf.	gan gara	‘we have gone’
d.	torri zera	‘you have come’	cf.	gan zara	‘you have gone’
e.	in zezu	‘do it’	cf.	jan zazu	‘eat it’

In (11a), for example, the auxiliary *de* has an [e] mid vowel after the verb *torri*, ending in [i]. In contrast, the same auxiliary has a low [a] after the verb *gan*, whose last vowel is [a]. Similar patterns are found in other examples.

Hualde discusses this data in a chapter where his goal is to “explore the interaction between the phonological rules of Basque and different morphological operations, with the purpose of determining the role of the morphology in defining the domain of application of phonological processes.” (op. cit.:22) Hualde claims that morphemes in which /a/ is unaffected belong to different morphological strata from those in which /a/ undergoes LVA. However, it seems difficult to maintain this position given his characterization of the data.

(12) Exceptions to LVA (op. cit.:26–30):

- “Not all derivational suffixes present the same behaviour with respect to Low Vowel Assimilation.”
- “Verbal suffixes generally undergo assimilation.”
- “We need to examine now the application of the rule in conjugated verbal forms. Here, the situation is not uniform... [I]n quite a few conjugated forms the rule fails to apply.”
- “The forms that present the context for Low Vowel Assimilation, but, nevertheless, do not undergo the rule, have just one more irregularity that must be lexically marked.”
- “Certain auxiliary verbs also undergo assimilation... Other forms of the auxiliary, on the other hand, never undergo assimilation, including the other persons of the intransitive present indicative not mentioned above...”
- “We must conclude that only a few auxiliary forms can behave like clitics and thus undergo Low Vowel Assimilation.”

Our analysis of LVA is far simpler than Hualde’s account of the exceptional morphemes, given the large number of exceptional morphemes. For example, in contrast with (11), there are other auxiliary verb forms that do not exhibit LVA. The 1sg. *naiz*, the 2sg. informal *aiz* and the 2pl. form *zate* are all invariant, as seen in *torri naiz* [toři naiš] ‘I have come’. It is hard to imagine any account which would place non-alternating auxiliaries like *naiz* and alternating auxiliaries like those in (11) in different morphological strata. We propose that Baztan LVA is best analyzed as an instance of inalterability as prespecification. We posit an underlying three-way distinction between /a, e, A/, in which /A/ is [−ROUND, −HIGH] but unspecified for Low and BACK. It is not the case that the different auxiliary forms are assigned to different morphological strata; rather, some of them contain an underspecified /A/ which, because of its underspecification, is *mutable* via a unification rule, and others contain an fully-specified /a/ which is *inalterable* (with respect to this rule) because its richer specification causes unification failure. The rule is given below.

$$(13) \left[\begin{array}{c} -\text{HIGH} \\ -\text{ROUND} \end{array} \right] \sqcup \{-\text{Low}\} / [+ \text{HIGH}] \text{ } C_0 \text{ —}$$

The relevant portion of the Baztan phoneme inventory are the vowels /a, e, A, o/. Unification of /a/ with {–Low} is vacuous because of unification failure. Unification of /e, o/ is vacuous because they are already specified –Low. Only unification with /A/ is non-vacuous and produces raising.

Now, it is also necessary to ensure that this raised vowel surfaces as the {–BACK, –ROUND} vowel [e] and not, say [o], and that /A/s which does not undergo (13) surface as [a], but there are several ways to ensure that, and we leave off discussion of this issue for reasons of space. Crucially, though, rule (13) *appears* to have targeted only the underspecified segment even though LP’s notion of natural classes and feature-filling make it impossible to do so, since the smallest natural class containing /A/ contains all four non-high vowels /a, e, A, o/.

This example illustrates themes (9i) and (9iii). Hualde’s appeal to morphological strata to generate exceptions to LVA seems unworkable, and indeed, he provides no means to stratify morphemes except by considering whether or not they are exceptional with respect to LVA. We avoid this by positing an underspecified vowel /A/, by its nature mutable and thus a potential target for LVA, and contrast this with pre-specified, inalterable /a/, which is immune to LVA. Unification and its failure implement a feature-filling rule without the need for rule diacritics or morphological strata. LP’s adoption of Inkelas’s insight means that there are no actual exceptions to LVA in the narrow phonology of Baztan.

4 Spanish third conjugation metaphony

We now turn to an example that has traditionally been referred to as metaphony (Malkiel 1966), the so-called “raising” alternations between *e* and *i* in Spanish verbs of the third (–*i*–) conjugation, as in *pedí* ‘I asked’ vs. *pido* ‘I ask’ or *gemí* ‘I wailed’ vs. *gimo* ‘I wail’. There are over one hundred raising verbs, though most of these stems are prefixed forms of a few dozen distinct roots. Many other third conjugation verbs, like *vivir* ‘to live’ and *sumergir* ‘to submerge’ do not alternate—the former has invariant *i* and the latter has invariant *e*. One could imagine analyzing the “raising” verbs as involving suppletion between listed *e* and *i* allomorphs (e.g., *ped-* vs. *pid-*), but as Embick (2012:33) notes, such an analysis would have to make reference to a complex and highly disjunctive set of morphosyntactic contexts.

(14) Morphosyntactic contexts for *ped-/pid-* (after Embick 2012:33):

- a. *ped-*: 1pl./2pl. present indicatives, 1sg./1pl./2sg./2pl. preterites, all imperfectives, all futures, all conditionals
- b. *pid-*: 1sg./2sg./3sg./3pl. present indicatives, all present subjunctives, all imperfect subjunctives, 3sg./pl. preterites

There is no obvious way to treat the difference between (14a) and (14b) in terms of natural classes based on morphosyntactic features, so the grammar would necessarily contain two lists—a list of

which verb *roots* participate in the alternation, and a list of which verb *forms* take *i* vs. *e*.⁵ Embick concludes that “[a]n analysis that makes reference to morphosyntactic features thus looks very unpromising”, and he offers a morphophonological alternative. He notes that *e* alternants of raising verbs are found when the following syllable contains an *i*, and *i* alternants are found in all other cases. Following a suggestion of Harris (1969:110f.), Embick takes the alternating vowel to be /i/, and proposes the following rule to generate the *e* allomorphs.

$$(15) i \rightarrow e / \text{ — } C_0 i \text{ (condition: certain roots)}$$

Embick’s proposal clearly improves on the morphosyntactic analysis, because it requires only a single list of which verb roots are included in (15) and not a list of specific forms, as in (14): this “(morpho)phonological analysis of the Raising verbs is straight-forward, as long as it is acknowledged that the phonological process can be Root-specific.” Note that Embick has replaced the traditional “raising” with a rule of lowering, since he has underlying /i/ surface as [e]. Given our adherence to the substance-freeness of structural change (2), we have nothing to say about his use of a dissimilatory lowering rule rather than an assimilatory raising rule.

However, with our LP technology, we can go a step further than Embick and replace his rule (15) with a narrow phonological rule. We can provide an account that reduces all idiosyncrasy to the phonological representation of individual morphemes in the lexicon, the level of the grammar which is characterized by idiosyncratic representations of contrastive elements. Suppose that the alternating vowel in raising verb stems is underlyingly /I/, with the same featural specification as in (4)—i.e., it is a vowel underspecified with respect to HIGH but otherwise the same as non-alternating /i, e/. Then instead of a rule like (16), we can provide a purely phonological rule that makes no reference to the identity of particular verb roots.

$$(16) \begin{bmatrix} -\text{BACK} \\ -\text{LOW} \end{bmatrix} \sqcup \{-\text{HIGH}\} / \text{ — } C_0 \begin{bmatrix} -\text{BACK} \\ +\text{HIGH} \end{bmatrix}$$

This rule will map underspecified /I/ to [e] when there is an /i/ in the next syllable, and apply vacuously to /i, e/. LP is unconcerned, given (2), that the rule inserts $-\text{HIGH}$ in the context of a trigger that is $+\text{HIGH}$.

To derive words like *pido*, in which /I/ is not followed by /i/ and thus not targeted by (16), we posit the following context-free rule, which applies vacuously to the fully-specified /e, i/.

$$(17) [-\text{LOW}] \sqcup \{+\text{HIGH}\}$$

This analysis again illustrates themes (9i) and (9iii), and it is quite similar in spirit to the analyses by Inkelas and colleagues and to our analysis of Baztan. Whereas in Baztan, where unraised /A/ surfaces as /a/, it is necessary to ensure that underspecified /I/ vowels which do not undergo (16) surface as [i] and not [e], but this poses no particular difficulty given the default rule (17). Our contribution is to show that the operations of LP, combined with Inkelas’ notion

⁵Embick also notes that a stem suppletion account for this Spanish case would run counter to his theory of locality conditions on stem suppletion (Embick 2010).

of inalterability as prespecification, allow us to move raising verb allomorphy into the narrow phonology. Crucially, our purely phonological solution does not exact any price: we only use tools independently proposed to address unrelated phenomena. We can account for Spanish raising verbs in the narrow phonology without enriching our model.

5 Cervara feature-changing metaphony

Metaphonic patterns cannot all be framed as feature-filling, however, so we now provide an analysis that incorporates a subtraction rule. A particularly clear case is apparent in the metaphony of Cervara discussed by Danesi (2022:§7.2.2). We present a simplified versions of Danesi’s data, restricting ourselves to front vowel targets in order to focus on the salient properties of the LP analysis.

5.1 Metaphony in verbs

We first consider metaphony in Cervara verbs. Pretheoretically, the two mid vowels /ε, e/ raise and neutralize to high, tense [i] when the final vowel is high, tense /i, u/.

(18) Cervara verbs (op. cit.:514):

- | | | | | |
|----|-------|------------|-------|-------------|
| a. | 'vedo | 'I see' | 'vidi | 'you see' |
| | 'vede | 'he sees' | 'vidu | 'they see' |
| b. | 'meto | 'I reap' | 'miti | 'you reap' |
| | 'mete | 'he reaps' | 'mitu | 'they reap' |

We hypothesize that the vowels which surface as [e] in “unraised” form are underlyingly specified as {−LOW, −HIGH, +ATR}, so, /e/; whereas those that surface as [ε] in unraised form are underlyingly {−LOW, −HIGH} but underspecified for ATR, a vowel we denote as /E/. We first propose the following rule which changes /E/ to [e] before a tense, high vowel, /i/ or /u/.

$$(19) \left[\begin{array}{c} -\text{LOW} \\ -\text{HIGH} \end{array} \right] \sqcup \{+\text{ATR}\} / \text{--- } C_0 \left[\begin{array}{c} +\text{HIGH} \\ +\text{ATR} \end{array} \right]$$

This is yet another feature-filling rule implemented with unification—like (13) or (16)—that neutralizes the contrast between the two mid vowels by adding a +ATR specification in a particular context. This rule feeds the next step of the process, which raises mid vowels specified +ATR if they are followed by a +HIGH vowel. This is a feature-changing process, because the target class includes vowels that are specified −HIGH. In LP, feature changing is a two-step process, with subtraction followed by unification. We first need a rule to remove the −HIGH specification from the relevant mid vowels.

$$(20) \left[\begin{array}{c} -\text{LOW} \\ +\text{ATR} \end{array} \right] \setminus \{-\text{HIGH}\} / \text{--- } C_0 \left[+\text{HIGH} \right]$$

Note that this will not remove the $-HIGH$ specification from underlying /E/ that persists at this stage, because that vowel lacks $+ATR$, unless it was neutralized to [e] by rule (19). This rule then feeds a third rule which raises $+ATR$ vowels when followed by a $+HIGH$ vowel.

$$(21) \left[\begin{array}{c} -LOW \\ +ATR \end{array} \right] \sqcup \{+HIGH\} / _ C_0 [+HIGH]$$

The feature-changing conversion of mid to high vowels is thus accomplished by a subtraction rule (20) followed by a unification rule (21), illustrating theme (9iv).

5.2 Metaphony in nominals

There is more to metaphony in Cervara. A different pattern is found in nouns and adjectives (henceforth, the nominals). In nominals, the underlyingly tense /e/ vowel raises to [i] when followed by a high vowel, just as we saw in verbs in (18). However, unlike the situation in verbs, in nominals underlyingly /E/ just shifts to /e/, surfacing as $+ATR$ but remaining mid.

(22) Cervara nominals (op. cit.:508–509):

- | | | | | |
|----|------------|------------------------|------------|-------------------------|
| a. | frel'lenka | ‘vulva (vulgar)’ | fril'linku | ‘penis (vulgar)’ |
| b. | 'b:elle | ‘beautiful (fem. pl.)’ | 'b:eʌʌi | ‘beautiful (masc. pl.)’ |

The solution we propose is essentially a translation of Danesi’s proposal into LP. Our analysis reflects theme (9i) in that it provides a narrow phonological solution for allomorphy that superficially appears to require morphological conditioning, and it illustrates theme (9ii) in that it involves an extension of Inkelas’s notion of prespecification from targets to triggers. Instead of distinguishing between mutable versus inalterable potential targets via underspecification, we instead distinguish between *catalytic* and *quiescent* potential triggers.

This extension of Inkelas’s notions appears implicitly in the literature, but we suspect that the implications have been insufficiently appreciated simply because the approach is so simple. Inkelas’s distinction between two types of potential targets via underspecification, rather than morphophonological rules or suppletion, requires no enhancements to narrow phonological theory, beyond admitting underspecification (something Inkelas and colleagues argue clearly for). Therefore, it is important to note that this same idea can also be extended to differentiate segments that are potential rule *triggers*.

Danesi’s analysis of Cervara (§7.2.2) makes use of this idea to avoid the need for lexical or morphosyntactic conditioning on metaphonic triggers. His analysis makes use of a complex representational system that makes assumptions about the relationship between phonetics and phonology contrary to ours. However, our proposal is a mere restatement of Danesi’s analysis in LP.

Once again, for ease of exposition, we restrict our attention only to front vowel targets. Back vowel behavior is completely symmetrical, so we set aside BACK and ROUND in our discussion. As above, we hypothesize that vowels which are realized as [e] in “unraised” form are $\{-LOW, -HIGH, +ATR\}$ and those that are realized as [ɛ] in unraised form are $\{-LOW, -HIGH\}$ but not specified for ATR, so /E/. Consequently, there exists a natural class containing just /e/ using $[-LOW, -HIGH,$

+ATR], but there is no corresponding singleton natural class for /E/: the class [−Low, −HIGH] also contains /e/. This is a desirable result.

Danesi presents an analysis of Cervara with two distinct sets of high vowels. One set, found in nominal endings, triggers tensing of [E] but raising of [e]. The other set triggers tensing and raising of /E/, and raising of both underlying and derived +ATR [e]. We propose that the two triggering vowels, both of which are realized as [i], are underlyingly distinct, so this is a case of absolute neutralization. The 2sg. verb ending /-i/ is specified [−Low, +HIGH, +ATR]. The nominal ending, which triggers the less drastic alternations, is denoted by /-Y/ and is specified [−Low, +HIGH], without a specification for ATR.⁶ By the same reasoning given above, there exists a singleton natural class containing only /i/, but the smallest natural class containing /Y/ also contains /i/.

In nominals, −ATR mid vowels change to +ATR in the presence of /Y/, whereas +ATR vowels become high vowels. Therefore, it is crucial to order the /E/-to-[e] rule after the /e/-to-[i] rule, in counterfeeding order. Recall that in verbal forms, both mid vowels raise to [i]. We now incorporate the rules developed above for verbs into an account that also covers nominals.

The first step is rule (19), repeated below as (23). This rule neutralizes the distinction between /e, E/ in the presence of a following /i/. Crucially, the +ATR environment specification ensures that /Y/ does not trigger this rule.

$$(23) \left[\begin{array}{c} -\text{Low} \\ -\text{High} \end{array} \right] \sqcup \{+\text{ATR}\} / \text{--- } C_0 \left[\begin{array}{c} +\text{High} \\ +\text{ATR} \end{array} \right]$$

The next step raises all +ATR mid vowels followed by a +HIGH vowel. This will target underlying /e...i, e...Y, E...i/ but not /E...Y/ since /Y/ is unspecified for ATR and thus will not have triggered the preceding rule. This is a feature-changing process, and therefore it is accomplished by a rule of subtraction followed by a rule of unification.

$$(24) \left[\begin{array}{c} -\text{Low} \\ +\text{ATR} \end{array} \right] \setminus \{-\text{High}\} / \text{--- } C_0 \left[+\text{High} \right]$$

$$(25) \left[\begin{array}{c} -\text{Low} \\ +\text{ATR} \end{array} \right] \sqcup \{+\text{High}\} / \text{--- } C_0 \left[+\text{High} \right]$$

Then, underlying /E/ before /Y/ is specified +ATR.

$$(26) [-\text{Low}] \sqcup \{+\text{ATR}\} / \text{--- } C_0 \left[+\text{High} \right]$$

Note that the target of (26) is the natural class intensionally characterized so as to include /e, E/, but in practice it only applies to /E/ because an /e/ followed by /i, Y/ will already have been raised by (25). Similarly, the trigger for (26) is intensionally characterized so as to include /i, Y/, but it only applies to sequences with /Y/ because there are no more /E...i/ sequences at this stage of the derivation thanks to rules (23)–(25).

⁶Note that we differ from Danesi's notation in order to follow the convention of using capital letters to denote underspecified segments.

The final two rules are redundancy rules. Such rules have no special status in LP: they are simply context-free unification rules, as we saw in Baztan and Spanish. The first rule introduces a +ATR specification to high vowels not specified for ATR. While this is only needed for /Y/, recall that our logic requires that we intensionally target /i, Y/, even though unification applies vacuously to the former. It is also important to note that a vowel like /Y/, lacks an ATR specification but may still trigger insertion of +ATR. As we noted in (2), LP does not require a copy of a feature inserted or deleted by a rule to be present in that rule’s structural environment. The second rule introduces a –ATR specification for mid vowels not specified for ATR, causing remaining tokens of /E/ to surface as [ɛ].

$$(27) [+HIGH] \sqcup \{+ATR\}$$

$$(28) [-HIGH] \sqcup \{-ATR\}$$

These last two rules can be expressed as context-free because at this late stage of the derivation, only a subset of vowels lack an ATR specification.

The following example shows the derivation of the relevant sequences. We use /I/, as in Spanish, to denote a vowel, appearing only in intermediate levels of derivation in Cervara, specified +ATR but underspecified with respect to HIGH. The last two columns provide derivations for the mid vowels not occurring in the contexts for metaphonic alternations.

(29) Cervara derivations:

	<i>vr.b.</i>	<i>nom.</i>	<i>vr.b.</i>	<i>nom.</i>		
UR	e...i	e...Y	E...i	E...Y	e	E
Rule (23):			e...i			
Rule (24):	I...i	I...Y	I...i			
Rule (25):	i...i	i...Y	i...i			
Rule (26):				e...Y		
Rule (27):		i...i		e...i		
Rule (28):						ɛ
SR	i...i	i...i	i...i	e...i	e	ɛ

Our analysis of Cervara can be adapted to similar cases of metaphony, for example, in Salentino, where historical vowel mergers make the pattern a challenge for traditional phonological analysis. In Salentino adjectives, for example, the [-i] of the masculine plural and the [-u] of the masculine singular appear with a raised vowel, but the [-i] of the feminine plural does not correlate with raising of the preceding mid +ATR vowel.

(30) Salentino adjectives (Torres-Tamarit and Linke 2016:§2.2):

	sg.	pl.	
fem.	'fredda	'fredi	'cold'
masc.	'friddu	'friddi	

A similar analysis to the one we provide for Cervara is clearly feasible here. Apparently, the underlyingly distinct affixes in Salentino are not members of different syntactic categories, but express different gender features. In LP, it is possible to posit different phonological features for the two plural endings, and it is unclear how to constrain LP to prevent analyses of this type from being posited.

6 Conclusions

We have attempted to provide a consistent feature logic under an explicit set of assumptions, such as the use of binary features and the possibility of underspecification. Despite the consistency and explicitness, many open questions remain and alternative analyses are available. For example, one might posit that the Cervara phoneme we denote as /E/ lacks an SPREAD GLOTTIS specification rather than an ATR specification. Furthermore, phonologists or morphologists may be skeptical of our use of absolute neutralization of underlyingly distinct segments, like Cervara /i, Y/. We present two arguments against such skepticism.

We call the first argument the ‘black hole’ argument. Black holes cannot be directly observed since no information can escape them. Their existence can only be discerned from their effect on other objects. Since our proposals for distinct phonological entities, for example, two variants of *i* in Cervara, are based on their differential effects on other elements (for example, to what degree they mutate a nearby /E/), we argue that there can be no principled reason to reject such solutions but to accept the existence of black holes. We are simply using the evidence at hand to infer a distinction, and it is conceivable that someday our hypothesis could be confirmed or falsified by neurological evidence.

The second argument is of a more general kind. It appears to be the case that a theory that makes use of ordered rules, binary features and underspecification, unification and subtraction, and the principle of substance-freeness of structural change (2) has no obvious way to exclude the sort of analyses we have proposed. In this light, we conclude, not that the proposals here are true and correct, but simply that it is difficult to demonstrate they are invalid, since the components they are built out of are all themselves well-supported. Our analyses follow from the components of our theory, which has no way to rule out a phonological system like Cervara with two underlying sources for *i*. Thus absolute neutralization is not only not forbidden, it is expected. Criticism of our assumptions and logic is of course quite welcome, but the overall system is fairly simple and yields rich empirical coverage via its combinatorics. A simple model that generates vast phenomenological complexity is exactly what we want out of any linguistic theory:

As concepts and principles become simpler, argument and inference tend to become more complex—a consequence that is naturally very much to be welcomed. (Chomsky 1982:3)

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